```
import pandas as pd

data = {
    "orderid": [
        "ORD1001", "ORD1002", "ORD1003", "ORD1004", "ORD1005",
        "ORD1006", "ORD1007", "ORD1008", "ORD1009", "ORD1010",
        "ORD1011", "ORD1012", "ORD1013", "ORD1014", "ORD1015",
        "ORD1016", "ORD1017", "ORD1018", "ORD1019", "ORD1020"
    ],

    "date": [
        "2026-01-05", "06/01/2026", "2026-01-07", "08-01-2026",
        "2026-01-09", "2026/01/10", "11-01-2026", "2026-01-12",
        "13/01/2026", "2026-01-14", "15-Jan-2026", "2026-01-16",
        "2026-01-17", "2026-01-18", "19/01/26", "2026-01-20",
        "2026-01-21", "13/01/2026", "2026-01-23", "24-01-2026"
    ],

    "region": [
        "North", "South", "East", "West", "North",
        "South", "East", "West", "North", "South",
        "East", "West", "North", "South", "East",
        "West", "North", "South", "North", "East"
    ],

    "product": [
        "Laptop", "Headphones", "Keyboard", "Monitor", "Mouse",
        "Smartphone", "Tablet", "Laptop", "Headphones", "Monitor",
        "Keyboard", "Smartphone", "Tablet", "Mouse", "Laptop",
        "Headphones", "Monitor", "Keyboard", "Smartphone", "Tablet"
    ],

    "units": [
        3, 12, 8, 4, 15, 5, 6, 2, None, 7,
        10, 3, 4, 20, 1, 9, 5, 6, 2, 3
    ],

    "unitprice": [
        55000, 1500, 800, 12000, 500,
        25000, 18000, 55000, 1500, 12000,
        800, 25000, 18000, 500, 55000,
        1500, 12000, 800, 25000, 18000
    ],

    "discountpct": [
        10, 5, 0, 15, 5, 10, None, 8, 5, 10,
        0, 12, 5, 0, 10, 5, None, 5, 15, 10
    ],

    "channel": [
        "Online", "Retail", "Online", "Distributor", "Retail",
        "Online", "Retail", "Online", "Distributor", "Retail",
        "Online", "Online", "Online", "Retail", "Distributor",
        "Online", "Retail", "Online", "Distributor", "Online"
    ]
}
```

```
df = pd.DataFrame(data)
```

```
df.head()
```

	orderid	date	region	product	units	unitprice	discountpct	channel
0	ORD1001	2026-01-05	North	Laptop	3.0	55000	10.0	Online
1	ORD1002	06/01/2026	South	Headphones	12.0	1500	5.0	Retail
2	ORD1003	2026-01-07	East	Keyboard	8.0	800	0.0	Online
3	ORD1004	08-01-2026	West	Monitor	4.0	12000	15.0	Distributor
4	ORD1005	2026-01-09	North	Mouse	15.0	500	5.0	Retail

Next steps: Generate code with df | New interactive sheet

```
df.tail()
```

	orderid	date	region	product	units	unitprice	discountpct	channel
15	ORD1016	2026-01-20	West	Headphones	9.0	1500	5.0	Online
16	ORD1017	2026-01-21	North	Monitor	5.0	12000	NaN	Retail
17	ORD1018	13/01/2026	South	Keyboard	6.0	800	5.0	Online
18	ORD1019	2026-01-23	North	Smartphone	2.0	25000	15.0	Distributor
19	ORD1020	24-01-2026	East	Tablet	3.0	18000	10.0	Online

```
df.shape
```

```
(20, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   orderid      20 non-null     object
 1   date         20 non-null     object
 2   region       20 non-null     object
 3   product      20 non-null     object
 4   units        19 non-null     float64
 5   unitprice    20 non-null     int64
 6   discountpct  18 non-null     float64
 7   channel      20 non-null     object
dtypes: float64(2), int64(1), object(5)
memory usage: 1.4+ KB
```

```
df.isna().sum()
```

	0
orderid	0
date	0
region	0
product	0
units	1
unitprice	0
discountpct	2
channel	0

dtype: int64

```
df["date"] = pd.to_datetime(
    df["date"],
    format="mixed",
    dayfirst=True,
    errors="coerce"
)
```

```
df['date'].dtype
```

```
dtype('<M8[ns]')
```

```
df['units'].isnull().sum()
```

```
np.int64(1)
```

```
df["units"] = df["units"].fillna(df["units"].median())
```

```
df['discountpct'].isnull().sum()
```

```
np.int64(2)
```

```
df["discountpct"] = df["discountpct"].fillna(
    df["discountpct"].median()
)
```

```
df.isnull().sum()
```

	0
orderid	0
date	0
region	0
product	0
units	0
unitprice	0
discountpct	0
channel	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df['revenue'] = df['units']*df['unitprice']*(1-df['discountpct']/100)
```

```
df[['product','region','units','unitprice','discountpct','revenue']]
```

	product	region	units	unitprice	discountpct	revenue
0	Laptop	North	3.0	55000	10.0	148500.0
1	Headphones	South	12.0	1500	5.0	17100.0
2	Keyboard	East	8.0	800	0.0	6400.0
3	Monitor	West	4.0	12000	15.0	40800.0
4	Mouse	North	15.0	500	5.0	7125.0
5	Smartphone	South	5.0	25000	10.0	112500.0
6	Tablet	East	6.0	18000	6.5	100980.0
7	Laptop	West	2.0	55000	8.0	101200.0
8	Headphones	North	5.0	1500	5.0	7125.0
9	Monitor	South	7.0	12000	10.0	75600.0
10	Keyboard	East	10.0	800	0.0	8000.0
11	Smartphone	West	3.0	25000	12.0	66000.0
12	Tablet	North	4.0	18000	5.0	68400.0
13	Mouse	South	20.0	500	0.0	10000.0
14	Laptop	East	1.0	55000	10.0	49500.0
15	Headphones	West	9.0	1500	5.0	12825.0
16	Monitor	North	5.0	12000	6.5	56100.0
17	Keyboard	South	6.0	800	5.0	4560.0
18	Smartphone	North	2.0	25000	15.0	42500.0
19	Tablet	East	3.0	18000	10.0	48600.0

```
df['revenue'].describe()
```

	revenue
count	20.000000
mean	49190.750000
std	42125.569285
min	4560.000000
25%	9500.000000
50%	45550.000000
75%	70200.000000
max	148500.000000

dtype: float64

```
df['product'].value_counts()
```

product	count
Laptop	3
Headphones	3
Keyboard	3
Monitor	3
Smartphone	3
Tablet	3
Mouse	2

dtype: int64

```
revenue_by_region = df.groupby('region')['revenue'].sum().sort_values(ascending=False)

revenue_by_region
```

region	revenue
North	329750.0
West	220825.0
South	219760.0
East	213480.0

dtype: float64

```
units_sold_by_products = df.groupby('product')['units'].sum().sort_values(ascending=False)

units_sold_by_products
```

product	units
Mouse	35.0
Headphones	26.0
Keyboard	24.0
Monitor	16.0
Tablet	13.0
Smartphone	10.0
Laptop	6.0

dtype: float64

```
region_summary = (
    df.groupby('region')
    .agg(
        TotalRevenue = ("revenue","sum"),
        AverageOrderRevenue = ("revenue","mean"),
        OrderCount = ("orderid","count")
      )
    .sort_values("TotalRevenue",ascending=False)
)

region_summary
```

region	TotalRevenue	AverageOrderRevenue	OrderCount
North	329750.0	54958.333333	6
West	220825.0	55206.250000	4
South	219760.0	43952.000000	5
East	213480.0	42696.000000	5

Next steps: Generate code with `region_summary` New interactive sheet

```
highest_revenue_orders = (
    df.sort_values("revenue", ascending=False)
    [
        [
            'orderid',
            'date',
            'region',
            'product',
            'units',
            'unitprice',
            'discountpct',
            'revenue'
        ]
    ]
    .head(5)
)

highest_revenue_orders
```

	orderid	date	region	product	units	unitprice	discountpct	revenue
0	ORD1001	2026-01-05	North	Laptop	3.0	55000	10.0	148500.0
5	ORD1006	2026-01-10	South	Smartphone	5.0	25000	10.0	112500.0
7	ORD1008	2026-01-12	West	Laptop	2.0	55000	8.0	101200.0
6	ORD1007	2026-01-11	East	Tablet	6.0	18000	6.5	100980.0
9	ORD1010	2026-01-14	South	Monitor	7.0	12000	10.0	75600.0

Next steps: Generate code with `highest_revenue_orders` New interactive sheet

```
numeric_columns = ['units','unitprice','discountpct','revenue']
correlation_table = df[numeric_columns].corr()
correlation_table
```

	units	unitprice	discountpct	revenue
units	1.000000	-0.656662	-0.677010	-0.531819
unitprice	-0.656662	1.000000	0.537220	0.763241
discountpct	-0.677010	0.537220	1.000000	0.501242
revenue	-0.531819	0.763241	0.501242	1.000000

Next steps: Generate code with `correlation_table` New interactive sheet

```
units_revenue_correlation = df['units'].corr(df['revenue'])
units_revenue_correlation
```

```
np.float64(-0.5318193294168012)
```

```
df['month'] = df['date'].dt.month_name()
df['month']
```

	month
0	January
1	January
2	January
3	January
4	January
5	January
6	January
7	January
8	January
9	January
10	January
11	January
12	January
13	January
14	January
15	January
16	January
17	January
18	January
19	January

dtype: object

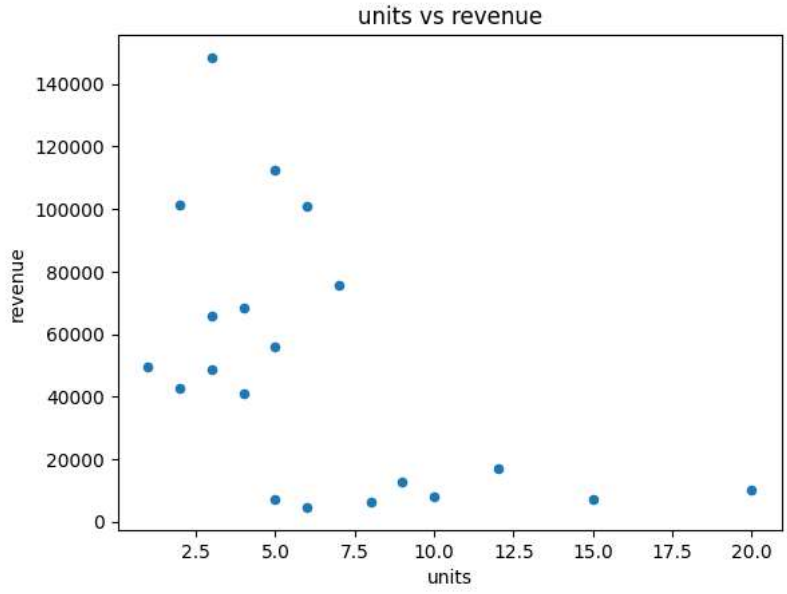
```
monthly_revenue = df.groupby('month')['revenue'].sum().sort_index()
monthly_revenue
```

month	revenue
January	983815.0

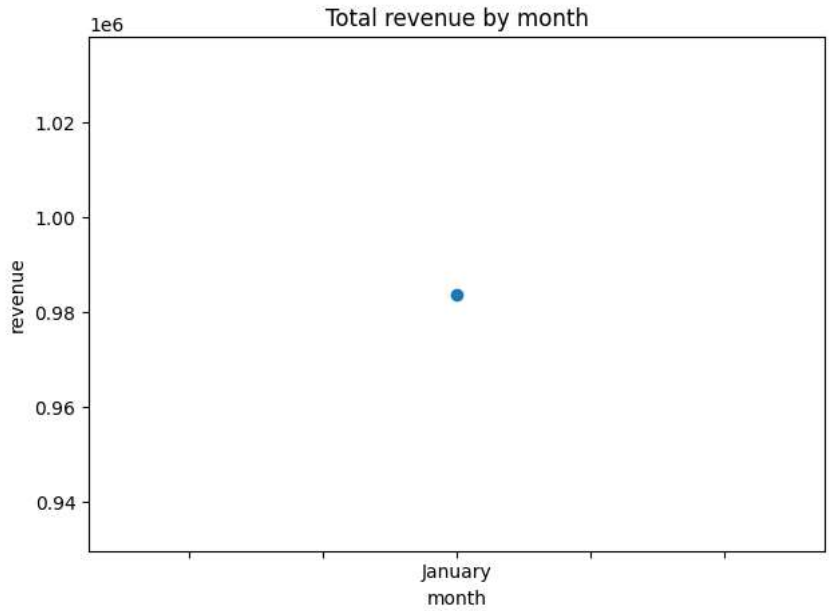
dtype: float64

```
import matplotlib.pyplot as plt
```

```
df.plot(
    kind="scatter",
    x="units",
    y="revenue",
    title="units vs revenue"
)
plt.show()
```




```
monthly_revenue.plot(
    kind="line",
    marker="o",
    title="Total revenue by month"
)
plt.xlabel("month")
plt.ylabel("revenue")
plt.tight_layout()
plt.show()
```

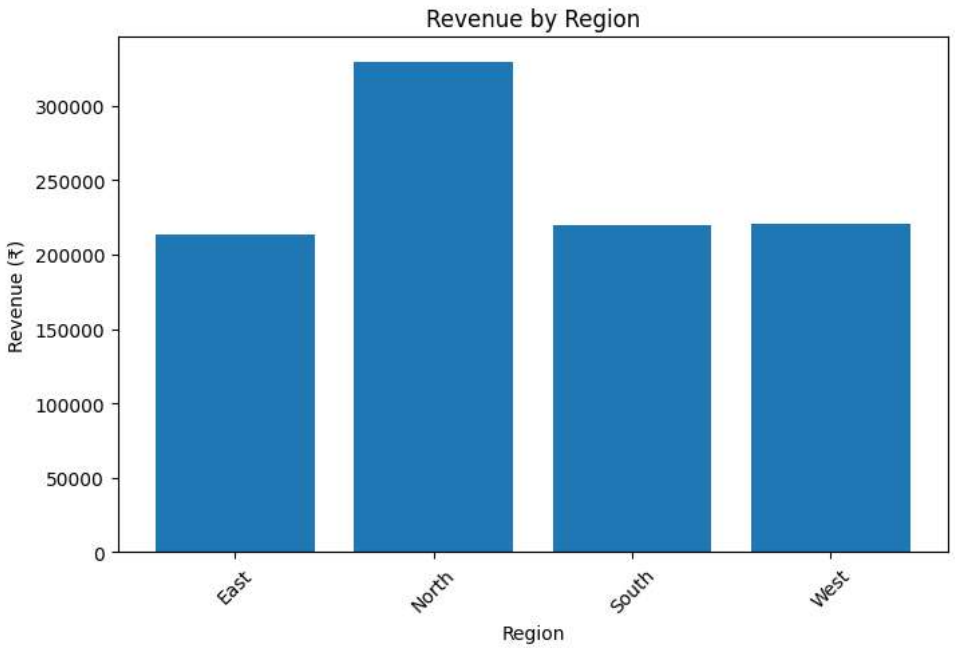



```
region_revenue = df.groupby("region")["revenue"].sum()

plt.figure(figsize=(8, 5))
plt.bar(region_revenue.index, region_revenue.values)

plt.title("Revenue by Region")
plt.xlabel("Region")
```

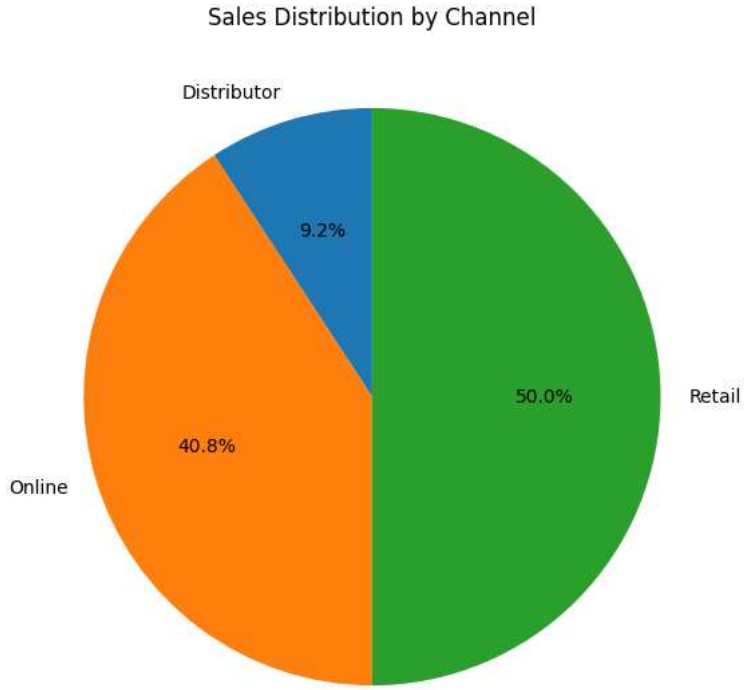
```
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=45)
plt.show()
```




```
channel_sales = df.groupby("channel")["units"].sum()

plt.figure(figsize=(7, 7))
plt.pie(
    channel_sales,
    labels=channel_sales.index,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Sales Distribution by Channel")
plt.show()
```

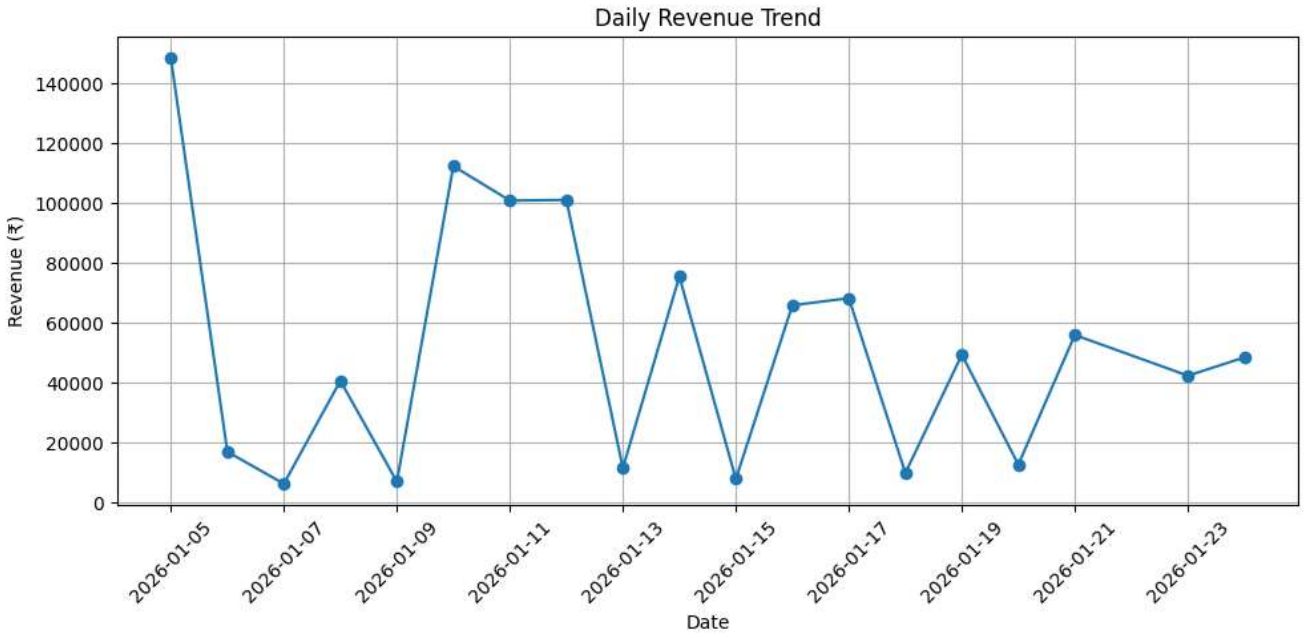


```
daily_revenue = df.groupby("date")["revenue"].sum()

plt.figure(figsize=(10, 5))
plt.plot(daily_revenue.index, daily_revenue.values, marker="o")

plt.title("Daily Revenue Trend")
plt.xlabel("Date")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=45)
plt.grid(True)

plt.tight_layout()
plt.show()
```




```
price_comparison = df.groupby("region")["unitprice"].mean()

print(price_comparison)

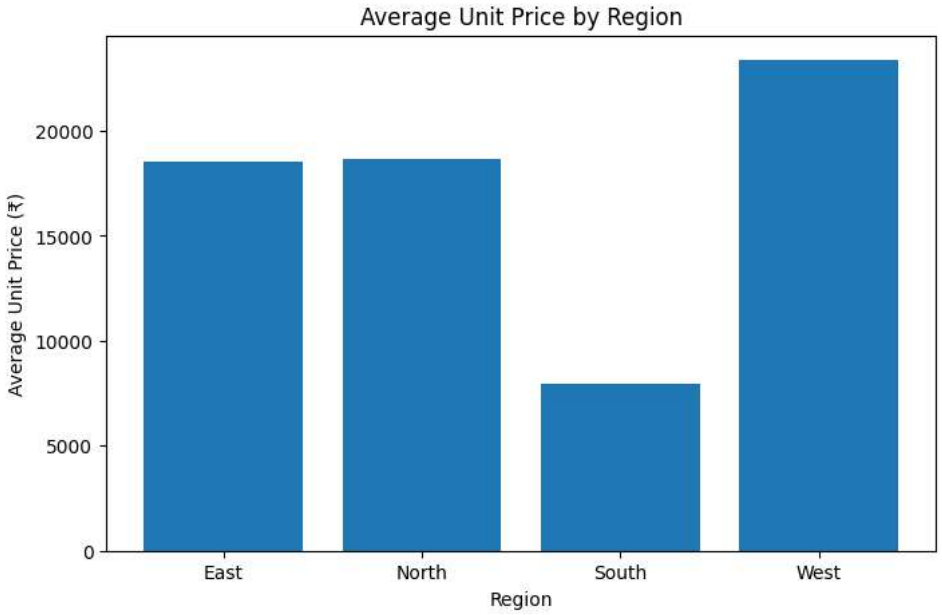
plt.figure(figsize=(8, 5))

plt.bar(price_comparison.index, price_comparison.values)

plt.title("Average Unit Price by Region")
plt.xlabel("Region")
plt.ylabel("Average Unit Price (₹)")

plt.show()
```

```
region
East     18520.000000
North    18666.666667
South     7960.000000
West     23375.000000
Name: unitprice, dtype: float64
```




```
region_product = df.pivot_table(
    index="region",
    columns="product",
    values="revenue",
    aggfunc="sum"
)

print(region_product)

region_product.plot(
    kind="bar",
    figsize=(12, 6)
)

plt.title("Product Revenue Comparison Across Regions")
plt.xlabel("Region")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=0)
plt.legend(title="Product")
plt.tight_layout()
plt.show()
```

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